

Figure 4.5: Krebs cycle

After citrate has been formed, the cycle machinery continues through seven distinct enzyme catalyzed reactions that produce in order isocitrate, α – ketoglutarate, succinyl CoA, succinate, fumarate, malate and oxaloacetate.

At the end of Krebs cycle, each pyruvic acid produces 2 CO_2, 1 ATP (substrate level phosphorylation), 3 NADH and 1 $FADH_2$. Then NADH and $FADH_2$ can be oxidized by electron transport chain to provide more ATPs.

4.6 Electron Transport Chain

An electron transport chain consists of a sequence of carrier molecules that are capable of oxidation and reduction. In Eukaryotic cell, the ETC is contained in the inner membrane of mitochondria or chloroplast membrane, whereas in prokaryotic cells, it is found in plasma membrane or cytoplasmic membrane.

The ETC is carried out through a series of electron transporters embedded in the

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